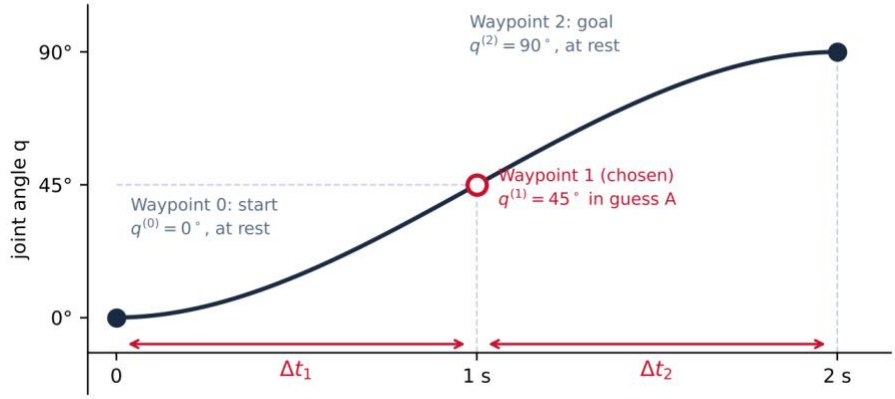
Example: One Joint, One Intermediate Waypoint

Setup

One joint rotates from $0°$ to $90°$, starting and ending at rest, with one intermediate waypoint in between. Speed limit $90°/s$, weight $w = 10^{-4}$. In the full problem each waypoint $q^{(k)}$ is a vector of 3 joint angles; here there is one joint, so it is a single number.

$$\mathbf{x} = \left[\, q^{(1)},\ \Delta t_1,\ \Delta t_2 \,\right] \in \mathbb{R}^3$$

The decision variables are the waypoint angle and the two segment durations.



Step 1: one cubic per segment

With τ the time since the segment began:

$$q_1(\tau) = a_1 + b_1\tau + c_1\tau^2 + d_1\tau^3 \quad (0 \le \tau \le \Delta t_1),$$
$$q_2(\tau) = a_2 + b_2\tau + c_2\tau^2 + d_2\tau^3 \quad (0 \le \tau \le \Delta t_2)$$

Velocity $\dot{q} = b + 2c\tau + 3d\tau^2$, acceleration $\ddot{q} = 2c + 6d\tau$, jerk $j = 6d$ (constant on each segment). That is 8 unknown coefficients.

Step 2: eight conditions fix the eight coefficients

Start	Waypoint	Goal
$q_1(0) = 0°$	$q_1(\Delta t_1) = q^{(1)},\ q_2(0) = q^{(1)}$	$q_2(\Delta t_2) = 90°$
$\dot{q}_1(0) = 0$	$\dot{q}_1(\Delta t_1) = \dot{q}_2(0),\ \ddot{q}_1(\Delta t_1) = \ddot{q}_2(0)$	$\dot{q}_2(\Delta t_2) = 0$

So every choice of $\mathbf{x}$ gives exactly one smooth trajectory.

Step 3: solve

With average slopes $s_1 = q^{(1)}/\Delta t_1$ and $s_2 = \left(90 - q^{(1)}\right)/\Delta t_2$, matching acceleration at the waypoint gives the waypoint velocity:

$$v_1 = \frac{3\,(\Delta t_2\, s_1 + \Delta t_1\, s_2)}{2\,(\Delta t_1 + \Delta t_2)}$$

and then

$$d_1 = \frac{v_1 - 2s_1}{\Delta t_1^2}, \qquad d_2 = \frac{v_1 - 2s_2}{\Delta t_2^2}$$

v_1 **is not a decision variable.** It is fixed by $\mathbf{x}$. Choosing it freely would let acceleration jump at the waypoint, which means infinite jerk.

Step 4: objective in the decision variables

$$J = T + w\int_0^T j\,(t)^2\,dt \;=\; \Delta t_1 + \Delta t_2 + 36\,w(d_1^2\,\Delta t_1 + d_2^2\,\Delta t_2)$$

The integral is exact because jerk is constant on each segment.

Step 5: evaluating one guess

Guess A: $\mathbf{x} = [\,45°,\ 1\text{ s},\ 1\text{ s}\,]$.

Quantity	Calculation	Value
Slopes s_1, s_2	$45/1,\ 45/1$	$45,\ 45$ °/s
Waypoint velocity v_1	$3(45+45)/(2\cdot 2)$	67.5 °/s
$d_1,\ d_2$	$(67.5-90)/1$	$-22.5,\ -22.5$
Jerk $6d_1,\ 6d_2$		-135 °/s³ on each segment
Objective J	$2 + 10^{-4}\cdot 36\,(22.5^2 + 22.5^2)$	$2 + 3.645 = 5.65$

The joint follows $q_1(\tau) = 67.5\tau^2 - 22.5\tau^3$, then $q_2(\tau) = 45 + 67.5\tau - 22.5\tau^3$:

Time	0 s	0.5 s	1 s (waypoint)	1.5 s	2 s (goal)
Angle	$0°$	$14.1°$	$45°$	$75.9°$	$90°$
Velocity (°/s)	0	50.6	67.5	50.6	0

Step 6: the optimizer adjusts x to lower J

Guess	$q^{(1)}$	$\Delta t_1,\ \Delta t_2$	T	Peak speed	J
A: starting guess	$45°$	1, 1 s	2.0 s	67.5 °/s	5.65
B: waypoint moved	$60°$	1, 1 s	2.0 s	80.4 °/s	12.13
C: faster	$45°$	0.6, 0.6 s	1.2 s	112.5 °/s (too fast)	48.08
D: best equal times	$45°$	1.45, 1.45 s	2.89 s	46.7 °/s	3.469
Optimum	$4.8°$ or $85.2°$	0.42, 2.44 s (or mirrored)	2.86 s	47.7 °/s	**3.434**

Moving faster cuts T but raises jerk; the optimum balances the two. The symmetric point D is a saddle, and the two mirror-image optima show the problem is **nonconvex**, so the solver needs several starting points. The full problem works the same way with 3 joints, 3 waypoints, and 13 variables.